

Jinkyounghwangbo

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EDUCATION

University of Illinois Urbana-Champaign

Master of Science in Computer Science

Aug 2026– May 2028 (Expected)

Illinois, USA

Seoul National University

Bachelor of Science in Industrial Engineering and Artificial Intelligence

Mar 2018–Aug 2022

Seoul, Republic of Korea

- Honor: Summa Cum Laude, GPA: 4.06/4.30, Rank: 3rd

Sejong Academy of Science and Arts

National STEM-gifted high school

Mar 2015–Jan 2018

Sejong, Republic of Korea

WORK EXPERIENCE

Samsung Electronics

Full Time Researcher @ Samsung Research, Mobile eXperience Business

Jul 2022–Jun 2026

Seoul, Republic of Korea

- Audio Eraser for Galaxy S Series 
 - Designed an on-device AI sound detection model working in real time, improving accuracy by over 15% while cutting computing cost by 10%
 - Formulated a method that analyzes both audio and video to automatically identify meaningful highlight moments, resulting in two patent filings [P.1, P.2]
 - Conducted research on audio-visual separation, using visual cues to isolate individual sound sources
- Acoustic Anomaly Detection for Samsung Home Appliances
 - Designed a training method that resolves subjective labeling, achieving 96.3% fault-detection accuracy – outperforming human experts (87.3%) in blind testing
 - Devised a data-efficient algorithm using <4 hours of training data, generating \$1.3M in estimated annual cost reduction and earning selection for Samsung Tech Fair 2024

Samsung Electronics

Research Intern @ Samsung Research

Summer 2021

Seoul, Republic of Korea

- Speech Bandwidth Extension with Hifi-GAN
 - Created a generative model extending speech audio bandwidth from 8 kHz to 16 kHz, improving perceptual quality scores by 31% across diverse speakers and conditions

Seoul National University

Research Intern @ Human Centered Computing Lab

Summer 2020

Gwanggyo, Republic of Korea

- ECG-based Heart Disease Prediction

PATENTS

[P.2] Jinkyounghwangbo, Hosang Sung, Seong-Hu Kim, et al. "Electronic Device and Method for Summarizing Video Data". KR Patent Application, P2025-0128738. Filed: Sep 10, 2025.

[P.1] JeongHoon Lee, Yeaseul Song, Jinkyounghwangbo, et al. "Electronic Device and Method for Generating Summarized Video Data". KR Patent Application, P2025-0115037. Filed: Aug 19, 2025.

HONORS AND AWARDS

- **Samsung Research Award (\$2,200 USD)** Dec 2024
Awarded project biannually for exceptional technical achievement across global research centers.
- **Merit-based Scholarship (Top 5%) @ Seoul National University** 2019–2022
Full amount (3 semesters), 50% (3 semesters), and 10% (1 semester) of tuition waiver.
- **Reporter Award @ Seoul National University** Nov 2019
- **4th Supercomputer Idea Contest (1st place, \$500 USD)** Oct 2016
- **3rd Supercomputer Idea Contest (1st place, \$700 USD)** Oct 2015

TEACHING / MENTORING EXPERIENCE

- **Teaching Assistant** Mar 2021–Jun 2022
Courses: Human Factors Engineering (406.304), Scientific Management (406.211)
- **Volunteer Social Service**
 - SNU Mentoring Program: Served online mentoring for high school students. Mar–Nov 2021
 - Teaching assistant for young adults with developmental disabilities. Spring 2019

ACTIVITIES

- **Tobigs: Machine Learning Study Group** Jan 2021–Jan 2022
 - Smootify [🔗]: Developed system for seamless song transitions, enabling continuous infinite streaming using music generation techniques with MuseGAN.
 - MPTI [🔗]: Developed chat service that allows users to interact with movie characters using DialoGPT and GPT-2.
- **Seoul National University News [🌐]** Apr 2018–Dec 2019
 - Head of Photography Department (1 semester) – Led department and managed visual content.
 - Head of New Media Team (1 semester) – Led newly established New Media team
 - Photojournalist (2 semesters) – Worked as photographer for university newspaper.

SKILLS

- **Tools and Frameworks:** PyTorch, TensorFlow, Tensorflow Lite, NumPy, Pandas, Matplotlib, SciPy, scikit-learn
- **Programming Languages:** Python, C/C++, Java, Kotlin, MySQL
- **Others:** Git, Github, Unit Testing, OOP, CI/CD